ULRR

A comparison of Irish and Swedish technology curricula: focus, form and function

Item Type	Article
Authors	Power, Jason Richard;O'Connor, Sean;Buckley, Jeffrey;Nordlóf, Charlotta;Hallström, Jonas
Citation	International Journal of Technology and Design Education
DOI	10.1007/s10798-026-10103-9
Publisher	Springer Nature
Rights	Attribution-NonCommercial-ShareAlike 4.0 International
Download date	2026-11-06 23:22:50
Item License	http://creativecommons.org/licenses/by-nc-sa/4.0/
Link to Item	https://hdl.handle.net/10344/31865

International Journal of Technology and Design Education
https://doi.org/10.1007/s10798-026-10103-9

A comparison of Irish and Swedish technology curricula: focus, form and function

Jason Richard Power[1] · **Sean O'Connor**[1] · **Jeffrey Buckley**[2,3] · **Charlotta Nordlöf**[3] · **Jonas Hallström**[3]

Received: 29 January 2026 / Accepted: 11 May 2026
© The Author(s) 2026

Abstract
Technology education varies significantly across national jurisdictions, reflecting diverse philosophical and historical trajectories. This study presents a comparative analysis of the intended technology curricula for lower secondary education in Ireland and Sweden. Using the three part heuristic framework for technology education, the research examines the distribution of learning outcomes across three categories which consist of technical skills, technological scientific knowledge, and socio ethical technical understanding. A qualitative content analysis was conducted on the subject specifications for the four Irish technology subjects, Applied Technology, Engineering, Graphics, and Wood Technology, alongside the single Swedish Technology subject. The findings reveal a marked divergence in pedagogical priorities between the two nations. Irish curricula are characterised by a high concentration of technical skills, particularly in Engineering and Graphics, where socio ethical considerations are notably minimal or absent. In contrast, the Swedish curriculum places a significantly higher emphasis on socio ethical understanding and the societal impact of technology, reflecting a more balanced epistemological profile. These results suggest a technical dominance within the Irish suite that potentially limits the development of holistic technological literacy. The paper concludes by discussing the implications of these findings for the ongoing redevelopment of the Irish Senior Cycle. It advocates for a more balanced integration of ethical and critical perspectives to ensure that students are prepared for the complexities of a contemporary technological society.

Keywords Technology syllabus · Technology education · Second-level education · Curriculum · Comparative study

✉ Jason Richard Power
Jason.power@ul.ie

[1] School of Education, Faculty of Education & Health Sciences, University of Limerick, Limerick, Ireland

[2] Department of Technology Education, Faculty of Engineering and Informatics, Technological University of the Shannon: Midlands Midwest, Athlone, Ireland

[3] Deptartment of Behavioural Sciences and Learning (IBL), Linköping University, Norrköping, Sweden

Published online: 19 May 2026

Introduction

Technology education has developed along markedly different trajectories across nations, shaped by distinct philosophical, policy and structural conditions (Benson & Lunt, 2011; Jones et al., 2013). Comparative studies have shown that ostensibly similar subjects diverge in rationale, content emphasis and assessment regimes, with implications for what students come to know and be able to do (Rasinen, 2003; Stronati, 2023). Critically examining curricula is therefore not a descriptive exercise but a means of producing analytic clarity about which features are national contingencies and which appear recurrent, of informing and calibrating reform efforts by exposing alternative design choices, and of providing interpretive context for differences in student outcomes across systems (Eurydice, 2022; Jones & de Vries, 2009). This paper contributes to that comparative endeavour by analysing and contrasting the intended technology curricula in Ireland and Sweden. For the purposes of this study the intended curriculum is characterised as the relevant official written national curriculum.

Curricular heterogeneity and subject identity

A widely cited publication by Jones et al. (2013) located the evolution of school technology education within emergent philosophical framings, international curriculum reform, and its shifting attachment to Science, Technology, Engineering, and Mathematics (STEM) policy. They show that the intended, enacted and attained curriculum is highly contingent, varying across countries, regions, schools and classrooms. More recent work suggests that this variance still exists in both curricular structure, future projections and how the field itself is conceptualised (Dakers, 2025; Doyle et al., 2019). Comparative handbooks likewise document substantial cross-national variation in curricular organisation (Benson & Lunt, 2011; de Vries, 2018; Jones & de Vries, 2009).

A recurrent concern is the unstable curricular status of technology. In some systems, STEM or Vocational Education and Training (VET) initiatives risk assimilating technology into broader frameworks, particularly where the subject lacks a defensible disciplinary identity (Jones et al., 2013; Williams, 2015). The United Kingdom experience further indicates that integration alone is insufficient. Durable reform requires conceptual clarity, robust initial and continuing teacher education, professional networks, aligned progression pathways and a maturing research culture (Barlex & Steeg, 2017; Beaumont & Steeg, 2024; Spendlove, 2017). Within Ireland, weak perceived connections to professional destinations have also been observed (Power & Buckley, 2024).

Surveying multiple national contexts, Banks and Williams (2022) argue that technology education should extend learners' understanding of technology's societal consequences, support the design of artefacts for defined needs, and engage students in authentic problem contexts relevant to global challenges. Yet, in many jurisdictions, practice remains patterned by the subject's craft-vocational heritage (Power & Buckley, 2024; Westerlund, 2023). Realising broader aspirations presupposes advances in research, curriculum and classroom practice.

This heterogeneity in curriculum structures and standards produces divergent conceptions of technology education, complicating comparison both internationally and within nations (Doyle et al., 2019). Unlike more established subjects, technology lacks a stable

philosophical core (Ihde, 2012; Schatzberg, 2018), which constrains analytical coherence. This is reflected in its vastly different realisations across nations (Rasinen, 2003), which stands in stark contrast to more consistent conceptualisations of science or maths which often follow established international guiding frameworks (Mullis et al., 2020). At the same time, this conceptual openness affords flexibility to adapt curricula to regional societal, environmental and industrial conditions (Banks & Williams, 2022).

Existing international comparisons

International comparisons of technology education for learners aged approximately 12 to 15 years typically focus on three dimensions. These dimensions consist of (1) the structural status of the subject within the curriculum, such as whether it is compulsory or elective and stand-alone or integrated within science, crafts, or informatics; (2) the intended content and epistemic profile of the subject, involving technical skill, design capability, technological literacy, and socio-ethical critique; and (3) the enacted pedagogical and assessment forms, such as iterative design projects, portfolio assessment, or paper-based tests (Rasinen, 2003).

A six-country analysis conducted by Rasinen (2003) demonstrated that, despite a shared policy vocabulary around technology, national interpretations diverge in the curricular rationale, knowledge structures, and assessment regimes attached to the subject. Stronati (2023) provided a broad overview of the differences in the design of secondary education across the European Union using data from the OECD. This report highlights significant divergence within the EU, noting that some jurisdictions retain an identifiable lower-secondary technology subject, while others disperse the content across craft, science, or digital competence strands, often with different assessment implications (Eurydice, 2022; Stronati, 2023).

Comparative research also examines specific thematic areas. Norström (2022) compared technological systems components in several national technology curricula and standards, including Sweden, England, Wales, Ireland, South Africa, New Zealand, Denmark, Finland, the USA, Scotland, and Australia. In a similar vein, Fredriksson et al. (2020) focused on differences in approaches to Education for Sustainable Development across Sweden and Japan. While not specifically addressing technology education, its approach to multinational comparisons of approaches to inclusion of sustainable development across curricula highlights the value of such efforts. Turja et al. (2009) provided an analysis of Early Childhood Education (ECE) technology curricula in six countries; Austria, Estonia, Finland, France, Germany and Scotland.

These comparative analyses are not merely descriptive; they yield several epistemic and policy benefits. First, they produce analytic precision by distinguishing culturally specific curricular choices from features that appear robust across systems. This helps to counter naive equivalence assumptions often present in single-country studies (Rasinen, 2003). Second, they provide policy calibration. By making visible the range of legitimate curriculum designs adopted elsewhere, international comparison functions as a check on insular reform trajectories and serves as an evidential basis for defending or revising national specifications (Eurydice, 2022). Third, they enable theory-to-evidence linkage. When cross-national differences in student outcomes, attitudes, or self-efficacy are observed, curricular divergence provides a principled explanatory context rather than leaving such variation uninterpreted (Eurydice, 2022; Rasinen, 2003; Stronati, 2023).

Irish and swedish comparisons

Sweden and Ireland were specifically selected for comparison because they serve as compelling case studies representing two fundamentally divergent educational philosophical foundations: The continental European Didaktik tradition and the Anglo-Saxon curriculum tradition (Tahirsylaj & Werler, 2021). While the Irish model is highly centralised and outcomes-based, prioritising the development of specialised makers through deep technical and craft-vocational mastery, the Swedish system is deeply rooted in the holistic concept of *Bildung*, which emphasizes teacher autonomy and the development of technologically literate citizens capable of social critique (Gleeson et al., 2025; Hopmann, 2007). This philosophical contrast makes the comparison highly meaningful, as international comparison of divergent systems functions as a check on insular reform trajectories and serves as an evidential basis for defending or revising national specifications. Analysing the Swedish approach provides an essential and useful reference point for Ireland by exposing alternative curricular design choices.

Lower secondary technology education in Ireland is delivered through a suite of four discrete subjects, which consist of Applied Technology, Engineering, Graphics, and Wood Technology. These subjects were recently redeveloped as part of the most significant national curriculum reform to date and are now situated within the Junior Cycle Framework (NCCA, 2024a). The Junior Cycle spans three years and typically serves learners aged 12 to 15 (DES, 2015; NCCA, 2024a). Its development involved a comprehensive review and redevelopment of all subjects across the curriculum. Upon completion of the Junior Cycle, the majority of students progress to the two-year Senior Cycle. This subsequent stage is currently undergoing a comparable reform process to ensure alignment with the Junior Cycle and to address the evolving needs of students as they transition out of post-primary education.

A central structural shift introduced through the Junior Cycle Framework is the replacement of traditional syllabi with specifications based on learning outcomes. Prior to this reform, technology syllabi articulated aims, objectives, and detailed lists of knowledge and skills. In contrast, the revised technology specifications now express intended learning through learning outcomes. The NCCA (2019, p.6) defines learning outcomes as statements in curriculum specifications to describe the knowledge, understanding, skills, and values students should be able to demonstrate after a period of learning. Learning outcomes act as the organising structures for planning, teaching, and assessment. This is then operationalised through the use of learning intentions and success criteria by teachers (NCCA, 2019). Given that learning outcomes now exert a defining influence on both what is taught and how it is enacted in classrooms, a close examination of their nature, emphasis, and distribution is both necessary and consequential.In Sweden, technology education is provided through a single subject in compulsory school, simply named Technology (Skolverket, 2022b). It is a compulsory subject for all students in grades 1 to 9, which encompasses ages 7 to 15. However, it remains one of the smaller subjects in regard to allocated teaching time in the curriculum, comprising a total of 200 teaching hours over the nine-year period While mathematics, one of the largest subjects, has just above 1200 teaching hours. While technology has been a part of the compulsory school curriculum since 1980, the subject has evolved significantly through successive reforms. Previously, it was closely connected to natural science subjects, but it has functioned as a standalone subject since 1994.

The primary aim of the subject is for students to develop "knowledge of the technology all around us" (Skolverket, 2022b, p. 274) and prepare them for life in a technological world, while also enabling them to pursue technical studies at higher levels. The curriculum introduces the subject with a general aim that culminates in three specific goals to provide students with the necessary foundations to develop (p. 274):

- the ability to reflect on different choices of technological solutions, their consequences for the individual, society and the environment, and how technology has changed over time,
- knowledge of technological solutions and how constituent components work together to achieve suitability and function, and
- the ability to perform technology development and design work.

The syllabus is structured into three core content areas. These include technology, people, society, and the environment; technological solutions; and working methods for developing technological solutions (Skolverket, 2022b).

In Sweden, curriculum design is heavily influenced by the German Didaktik tradition. In this curriculum approach, teachers are viewed as co-designers of the curriculum which means that they actively interpret and adapt the curriculum based on their students and the context. Consequently, the written Swedish national curriculum is by design broader, less prescriptive, and with more freedom given to the teachers. Thus, it is also less detailed and standardised compared to in school systems that are more centralised, and the depth and specificity is not always fully visible in the written curriculum itself; it is instead thought to be developed through teachers' professional practice in the classroom, for example (Clucas and Sjöström (2025).

A framework for technology curricula examination

Nordlöf et al. (2022) advance a heuristic framework for technology education grounded in established knowledge traditions. They note that technology curricula typically draw from craft, engineering, and the humanities and social sciences. A knowledge tradition refers to a community of practice in which shared skills and understandings evolve through both formal rules and informal, situated norms. These traditions overlap rather than form discrete epistemic categories (Håkanson, 2010).

The framework is presented as an epistemological tripod comprising three interdependent pillars: (1) technical skills, rooted in craftsmanship and validated through experience and performance (Ropohl, 1997; Vincenti, 1990), (2) technological scientific knowledge, associated with engineering and expressed through models, testing, standards and formal reasoning (Hansson, 2013; Norström, 2014), and (3) socio-ethical technical understanding, which examines technology in society using concepts and methods from the humanities and social sciences (de Vries, 2005, 2016).

Technical skills emphasise doing, including making, sketching, joining and repairing, with functionality prioritised over formal explanation. Technological scientific knowledge foregrounds explanation and prediction through analysis, calculation, documentation and modelling, typically learned through formal engineering education. Socio-ethical technical understanding focuses on technology as a human and societal phenomenon, engaging stu-

dents in description, historical tracing, ethical analysis and evaluation from sustainability or justice perspectives. Socio-ethical technical understanding is a vital component of a modern curriculum because it provides the critical framework necessary for technological literacy, allowing students to move beyond being mere users to understanding the complex relationship between technology, humanity, and the environment (de Vries, 2016). Adequate development of socio-ethical technical understanding empowers learners to evaluate the ethical, political, and sociological implications of artifacts, fostering the critical agency required for informed and responsible citizenship (Mathisen & Johansen, 2025).

Without adequate representation of this understanding, curricula fail to prevent the "black box syndrome," where individuals focus only on technical inputs and outputs while remaining blind to the power structures and intentions embedded within technology (Cederqvist, 2021). The ultimate consequence is a society of passive, uncritical consumers who are unable to evaluate how technical arrangements shape public life or foresee the long-range toll of technological innovation on human freedom (Winner, 2017). Together, the three pillars provide an organising schema for the heterogeneous knowledge claims that characterise school technology education (Nordlöf et al., 2022).

Aim of the study

This study compares the learning outcomes specified in Irish Junior Cycle technology subjects and the Swedish compulsory school technology subjects. The comparison is structured using the three-part heuristic framework for technology education developed by Nordlöf et al. (2022). By applying a common analytical lens to both national documents, the study seeks to characterise how the two systems distribute emphasis across different forms of technological knowledge and to identify where there may be scope for curricular augmentation or refinement.

To achieve this aim, the following research questions have been formulated to guide the investigation:

1. To what extent do the epistemological profiles of the Irish and Swedish technology curricula align with the three categories of the heuristic framework, consisting of technical skills, technological scientific knowledge, and socio ethical technical understanding?
2. What are the implications of the identified curricular focus for the development of holistic technological literacy and the ongoing redevelopment of the Irish Senior Cycle technology subjects?

These questions establish a roadmap for the subsequent analysis and ensure that the investigation addresses the specific gaps identified in international technology education literature (Rasinen, 2003; Stronati, 2023). Specifically, these works highlight a lack of analytical precision in characterising technology education's fragmented subject identity and the persistent tension between facilitating deep technical specialisation versus ensuring the curricular breadth required for holistic technological literacy. This study addresses these gaps by examining nationally specific curricular choices and the robust epistemological features necessary for a modern, balanced curriculum. By explicitly addressing these inquiries, the study aims to provide an evidential basis for future curricular recalibration, within the Irish context as it transitions toward a new Senior Cycle framework, within the Swedish con-

text as it begins a review effort considering the role of technology, and for more disparate international contexts seeking to further develop technology curricula. Such a comparison is essential for understanding how culturally specific curricular choices shape what students come to know and be able to do in a technological society (Nordlöf et al., 2022).

Methodology

Research design

This study utilised a qualitative content analysis to facilitate a detailed examination of the learning outcomes specified within the curriculum documents of Ireland and Sweden. The documents to be reviewed have been determined by how each country defines technology education. Qualitative content analysis was selected because it provides a systematic method for interpreting the meaning, structure, and content of specialised subject specifications (Forman & Damschroder, 2007). The analytical process was guided by the three part Heuristic Framework for Technology Education, which allowed for the classification of learning outcomes into categories of technical skills, technological scientific knowledge, and socio-ethical technical understanding (Nordlöf et al., 2022).

From an ethical perspective, the research was conducted as a form of documentary analysis. It drew exclusively on publicly available national curriculum documents provided by the National Council for Curriculum and Assessment in Ireland and the National Agency for Education (Skolverket) in Sweden. Consequently, no human participants were involved, and all primary sources have been appropriately acknowledged and attributed throughout the study. This approach ensured that the data collection was transparent and reproducible, adhering to the standards of academic integrity required for comparative curricular research.

Reflexivity

Given the interpretive nature of qualitative curriculum analysis, the research team engaged in a rigorous process of reflexivity and bracketing to ensure the credibility and trustworthiness of the findings (Moser & Korstjens, 2018). Reflexivity involves a critical self-examination of the researchers' own backgrounds and positions, acknowledging how these may influence the production of knowledge (Yip, 2024). In this study, reflexivity was operationalised through a collaborative, team-based approach designed to navigate the complexities of cross-national comparative research.

The research team was purposefully composed of five scholars with expertise in technology education across two distinct national contexts: Ireland and Sweden. This composition, comprised of three researchers based in Irish higher education institutions and two based in a Swedish university, allowed the team to address potential "methodological nationalism" (Phillips & Schweisfurth, 2014) by integrating both insider and outsider perspectives (Kelly, 2014). The Swedish members provided essential cultural and linguistic nuance when interpreting the Swedish curriculum, particularly regarding the *Didaktik* tradition, while the Irish members offered a comparative lens prompting deeper interrogation of assumptions inherent in both systems.

To manage potential bias, the team employed bracketing, a process of identifying and temporarily setting aside a priori assumptions and professional predispositions (Sorsa et al., 2015). Throughout the coding process, a recursive peer-debriefing strategy was used. Initial interpretations of the Swedish curriculum by the Irish-based researchers were systematically cross-verified and challenged by the Swedish-based researchers (Barry et al., 1999). This dialogic process ensured that the findings were grounded in the documentary evidence rather than the researchers' national or personal biases, thereby enhancing the overall rigor and transparency of the study.

Data collection

Learning outcomes from the four Junior Cycle Irish technology subject specifications (Department of Education & Skills, 2018a, 2018b, 2018c, 2018d) were categorised. Each specification presents a subject-specific rationale and explicitly positions the subject within the broader Framework for Junior Cycle (NCCA, 2024a), which defines the system-wide educational aims. The Framework sets out 24 Statements of Learning and associated key skills that all schools must address across students' programmes. The learning outcomes listed in the subject specifications served as the primary units of analysis, as they define the expected student learning. Claimed cross-curricular links and stated subject aims were also examined to provide contextual insight.

The Swedish curriculum for the subject of technology (Skolverket, 2022b) includes aim, core content, and grading criteria. The core content listed in the section for years 7 to 9 (ages 13 to 15) is divided into three sections, which include what must be taught during these school years. These sections of the curriculum served as units of analysis. As the curriculum is relatively concise with few examples, the supplementary document *Comments on the technology curriculum* (Skolverket, 2022a) also has been used to support the analysis. The commentary material complements the curriculum, providing context and depth.

Category definition

Categories were derived from the three-part heuristic framework for technology education proposed by Nordlöf et al. (2022). As they state, the framework is intended as a tool for analysing and comparing technology curricula, and despite national variation, the core content of technology education can be meaningfully reduced to these three categories (p. 1595). Each learning outcome was therefore coded as belonging to one of the following: (1) Technical Skills, (2) Technological Scientific Knowledge, or (3) Socio-ethical Technical Understanding.

Coding and analysis

The subject specifications/syllabi were reviewed in full and deductively coded using the predefined categories. 2 authors independently coded the curriculum documents. Initial coding was completed by an author with extensive experience in the relevant national specification/syllabi. A researcher based in the Republic of Ireland completed the initial coding of the Irish Curricula in its entirety. Similarly, a Swedish researcher with in-depth knowledge of the Swedish curriculum reviewed the relevant document and coded it in its entirety.

Initial coding was guided by the definitions and examples contained within the overarching Tri-Heuristic Framework (Nordlöf et al., 2022). Coding was then reviewed by each member of the authorship team independently. Discrepancies were resolved through discussion until agreement was reached amongst the entire authorship team. Category frequencies were calculated to inform qualitative interpretation, with the understanding that the number of learning outcomes is not a proxy for enacted pedagogical time or depth of learning. The qualitative analysis focused on interpreting the significance and contextual role of each category within the curriculum.

Findings

The analysis of learning outcomes (LOs) within the technology curricula of Ireland and Sweden, conducted using the Heuristic Framework for Technology Education, reveals significant differences in the focus and pedagogical priorities of each system. Learning outcomes were categorised into three primary categories: *Technical Skills*, *Technological Scientific Knowledge*, and *Socio-ethical Technical Understanding*.

Quantitative distribution of learning outcomes

The quantitative data highlights a distinct divergence between the Irish suite of technology subjects and the Swedish technology curriculum regarding the distribution of these dctaegories (Table 1).

Technical skills and the Irish suite

In the Irish context, all four subjects exhibit a dominant focus on Technical Skills, which account for the majority of learning outcomes in three out of the four subjects. Engineering shows the highest concentration in this category at 64.9%, closely followed by Graphics at 59.5%. This emphasis aligns with the stated aims of the Irish Junior Cycle Technology suite, which prioritises active learning environments where students can “advance their ideas from conception to realisation". For Engineering specifically, the focus is explicitly on “goal-oriented problem solving for the manufacture of products," with a heavy weighting towards efficiency and accuracy (see Fig. 1).

Table 1 Percentage distribution of learning outcome categories

	Wood Technology	Applied Technology	Engineering	Graphics	Swedish Technology
Technical Skills	47.4%	55.6%	64.9%	59.5%	23.5%
Technological Scientific Knowledge	36.8%	29.6%	32.4%	40.5%	47.1%
Socio-ethical Technical Understanding	15.8%	14.8%	2.7%	0.0%	29.4%

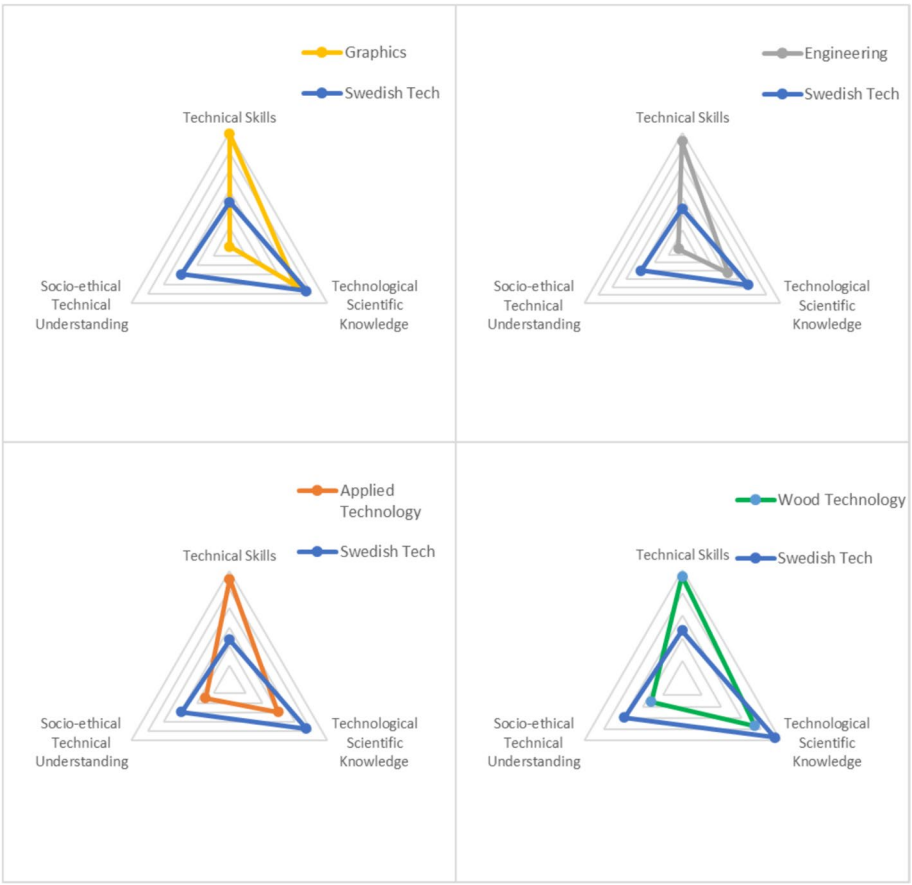


Fig. 1 Comparison of Irish technology subjects and Swedish technology subject

Contrast with Swedish curricular focus

Conversely, the Swedish technology curriculum demonstrates a more balanced distribution that prioritises Technological Scientific Knowledge (47.1%) and Socio-ethical Technical Understanding (29.4%) over direct technical mastery, Technical Skills (23.5%). The Swedish model places nearly double the emphasis on socio-ethical considerations compared to the most 'ethical' Irish subject, Wood Technology (15.8%). This suggests a curriculum designed to foster a broader understanding of technology's role in society rather than one focused primarily on the "how-to" of manufacture and design.

Disparities in socio-ethical understanding

There is a notable disparity in the inclusion of Socio-ethical Technical Understanding across the Irish subjects. While Wood Technology (15.8%) and Applied Technology (14.8%) incorporate outcomes related to the "impact of technology" and the "relationship with nature", Graphics (0.0%) and Engineering (2.7%) almost entirely omit these considerations. In

Graphics, the curriculum is instead positioned as the "underpinning language of the technology disciplines," focusing primarily on communication through various media.

Technological scientific knowledge consistency

Despite the variations in other categories, Technological Scientific Knowledge remains a consistent pillar across both nations, ranging from 29.6% in Applied Technology to 47.1% in the Swedish curriculum. This suggests that while pedagogical approaches to practice and society and ethics vary, the theoretical underpinnings required to understand technological design and technological systems remain a core component of technology education internationally.

Discussion

The comparative analysis of Irish and Swedish technology curricula through the application of the Heuristic Framework reveals fundamental differences in how these nations conceptualise the 'epistemological tripod' of technology education. According to Nordlöf et al. (2022), this framework consists of three mutually supporting legs: Technical Skills (the 'knowing-how' of practical manipulation), Technological Scientific Knowledge (the 'knowing-that' of rules, laws, and systemic functions), and Socio-ethical Technical Understanding (the 'knowing-why' and 'knowing-for-whom' regarding the impact and purpose of technology).

The practical-technical dominance in the Irish suite

The findings indicate that technology education in Ireland remains deeply rooted in a tradition of craft-based mastery and practical realisation. Across the four Irish subjects, Technical Skills are the primary focus. This weighting reflects the stated rationale of the Irish Engineering curriculum, which emphasises "goal-oriented problem solving for the manufacture of products" and the "efficient and accurate" use of materials.

Similarly, Wood Technology and Applied Technology exhibit high levels of technical skill requirements. The Wood Technology specification aims to develop students' ability to "apply the principles of design and realization" through a "project-led approach". While this focus ensures students are "resourceful and confident learners" in a workshop setting, it reinforces a subject identity built on 'knowing-how' rather than a broader critical understanding of technology as a social force.

These findings directly address the research aim of characterising national epistemological profiles, revealing that the Irish offering prioritises the technical depth, focused on material manipulation and functional accuracy. This research extends existing literature, such as that by Rasinen (2003) and Stronati (2023), by demonstrating that even modernised, design-focused specifications can remain patterned by traditional identities as pathways to technical vocations. A new insight contributed by this study is the identification of a socio-ethical vacuum in technically dense subjects like Engineering and Graphics, where the omission of social impact outcomes risks presenting technology to students as a value-neutral tool. For stakeholders, these findings highlight a distinct focus in learning outcomes that is at

odds with subject rationales contained within specification documents. This lack of internal consistency suggests that this is not a conscious curriculum design choice and highlights the need for external frameworks and benchmarks as curriculum evaluation mechanisms.

The socio-ethical disparity

A significant point of divergence is the treatment of Socio-ethical Technical Understanding. In the Swedish curriculum, nearly one-third (29.4%) of all learning outcomes are dedicated to this category reflecting a view of technology as a "social phenomenon" that requires students to evaluate its relationship with nature and society.

In contrast, the Irish subjects demonstrate what might be termed a 'socio-ethical vacuum', particularly in Graphics and Engineering. The Graphics specification defines itself as the "underpinning language of the technology disciplines" and focuses almost exclusively on communication and spatial reasoning. While these are essential technical and scientific competencies, the omission of outcomes related to the environmental or ethical impact of design choices further suggests that technology is presented to Irish students as a value-neutral tool. This lack of critical engagement may hinder the development of a holistic Technological Capability, which requires an understanding of how technology shapes and is shaped by human values (de Vries, 2016).

Socio-ethical technical understanding includes sociological, historical, philosophical, political, and ethical perspectives on technology in society, and how it can be understood and evaluated. The importance of these perspectives has been highlighted in historical, philosophical, and educational research in the last decades alongside the evolution of technology education (Hallström, 2025; Nye, 2006), so even though it does not concern technological knowledge per se, these perspectives constitute pivotal knowledge components and traditions necessary to understand technology itself. Ropohl (1997) designates this "socio-technological understanding", based on what Mitcham (1994) calls a "humanities philosophy of technology". It is commonly thought that this understanding should also be part of present-day technology curricula (Nordlöf et al., 2022) for students to gain a fuller understanding of technology.

Convergence in scientific underpinnings

Despite these differences, Technological Scientific Knowledge serves as a common epistemological foundation across both nations. This category, covering the scientific principles, functionality, and systemic rules of technology, remains relatively consistent.. For instance, Applied Technology focuses on principles and concepts such as electronics, mechanics, and control systems. The consistency of this category suggests that both jurisdictions recognise that modern technical competence requires a foundation in theoretical knowledge (knowing-that) to complement practical skill (knowing-how).

Subject identity: the tension between technical depth and curricular breadth

The structural delivery of technology education in Ireland through a suite of four discrete subjects presents a significant contrast to the single integrated subject model found in Sweden. This Irish approach appears to facilitate a high degree of technical specialisation and

craft based mastery. As indicated in the findings, Irish subjects such as Engineering and Graphics demonstrate a heavy weighting toward technical skills. This focus aligns with the traditional identity of these subjects as pathways to technical vocations or engineering disciplines, where the ability to realise a design and manipulate materials is considered paramount (Power & Buckley, 2024).

However, this structural depth in technical domains appears to come at the expense of curricular breadth, particularly regarding Socio ethical Technical Understanding. While the Swedish model allocates nearly 30% of its learning outcomes to the social and ethical implications of technology, the Irish subjects present few, or in some instances no, learning outcomes that address this area. This disparity suggests that the Irish suite model may inadvertently create curricular silos. In such a system, technical competencies are often isolated from the critical frameworks required to evaluate their impact on society and the environment.

As Nordlöf et al. (2022) argue, a robust technology education requires a balanced tripod of technical, scientific, and socio ethical knowledge. When these elements are fragmented across different subjects, students may struggle to develop a holistic understanding of technology as a complex and value laden human activity. The lack of socio ethical outcomes in the more technically dense Irish subjects like Engineering and Graphics reinforces a narrow perception of these fields. If a subject focuses almost exclusively on the how of manufacture, without the why of ethical consequence, it risks presenting technology as a value neutral tool (Nordlöf et al., 2022).

In contrast, the Swedish curriculum treats technology as a social phenomenon from the outset. This ensures that even at a lower secondary level, students engage with the relationship between technology, people, society, and the environment (Skolverket, 2022a, 2022b). This comparison highlights a fundamental tension in curriculum design. The Irish system prioritises the development of specialised makers and designers, while the Swedish system prioritises the development of technologically literate citizens capable of critique (Stronati, 2023). While the Irish suite provides students with a high level of agency in choosing their technical specialisation, the resulting silos may hinder the development of the broad critical perspective necessary to navigate a modern technological world. Future reforms in the Irish context, particularly within the Senior Cycle, should consider how to bridge these silos to ensure that technical expertise is supported by a strong ethical and social foundation.

Implications for senior cycle reform in ireland

The results of this analysis provide a critical evidence base for the current redevelopment of Senior Cycle technology subjects in Ireland (NCCA, 2024b). The identified concentration of technical skills at the expense of socio-ethical technical understanding suggests that the current Junior Cycle framework may be leading students to develop an incomplete form of technological literacy. As the National Council for Curriculum and Assessment moves forward with the largest reform of upper secondary education in decades, it is worth considering whether this distinct focus is an intentional design curricular choice. These findings also highlight an opportunity. The development of Senior Cycle subjects, which serve students in the last two years of their second level education, has the potential to adopt a different focus. When considering second level technology as a whole, this could result in a more holistic learning experience for students and align more readily with the Junior Cycle

(NCCA, 2024a) and Senior Cycle Frameworks (NCCA, 2024b) that guide the development of all students.

To address the socio-ethical vacuum found in subjects like Engineering and Graphics, new specifications should move beyond the mere recognition of technological impact and instead require active critical evaluation. One practical recommendation is the integration of life cycle thinking and ethical design criteria into the project based components of the subjects. Instead of focusing solely on the efficiency and accuracy of a manufactured product, learning outcomes should require students to assess the social and environmental consequences of their technical choices (Nordlöf et al., 2022). This would ensure that the knowing-why of technology is given equal weight to the knowing-how of manufacture.

Moreover, the Swedish example demonstrates that a single subject can successfully integrate scientific principles with social critique without sacrificing technical relevance. However, this inevitably leads to a trade off in depth versus span. While Ireland maintains a suite of subjects, the Senior Cycle reform offers a chance to harmonise these disciplines by introducing common socio-ethical themes such as sustainable development, data ethics, and the history of technological innovation (Power & Buckley, 2024). The current suite of subjects in Ireland provides an opportunity to take a different focus across the subjects. Where one subject is focused on technical skill development, another could place a heavier emphasis on socio-ethical understanding. In this manner both depth and breadth could be provided, albeit on a subject selection basis.

By making these dimensions explicit and assessable, the Irish system can evolve from a craft-centric tradition toward a more holistic model of technology education that aligns with the needs of a contemporary society. If these socio-ethical considerations remain absent from the intended curriculum, they are unlikely to be enacted in the classroom, potentially leaving students ill-equipped to navigate the complex moral landscape of modern engineering and design.

Limitations

While this study provides a systematic comparison of two distinct technology education traditions, several limitations must be acknowledged.

First, a primary limitation concerns the comparability of the units of analysis. The Irish curriculum is structured around "learning outcomes", prescriptive statements describing the knowledge, understanding, and skills students should demonstrate (NCCA, 2019, 2024a). In contrast, the Swedish curriculum is organised around "core content statements," which are traditionally broader and less prescriptive due to the Didaktik tradition's emphasis on the teacher as an autonomous "curriculum maker" (Bladh, 2020). This inherent difference in structural detail and philosophical intent may impact the direct comparability of category frequencies.

Second, this study focused exclusively on the intended curriculum as expressed in official national documents. Existing research distinguishes between the intended (official policy), the implemented (what is actually taught in the classroom context), and the attained curriculum (what students actually learn and achieve) (Wagner & Hastedt, 2022). As this is a documentary analysis, the findings do not account for classroom enactment or the "implicit curriculum," where the depth of a subject is often realised through a teacher's

professional judgment and pedagogical choices rather than being explicitly stated in curriculum documents.

Third, while category frequencies were calculated to provide a quantitative overview, the number of learning outcomes is not a direct proxy for enacted pedagogical time or the intellectual depth of student engagement. This is particularly relevant as the intended and implemented curriculum are highly contingent and can vary significantly across different jurisdictional regions and classrooms (Gleeson et al., 2025).

Conclusion

The application of the epistemological tripod framework has revealed a fundamental divergence in how Ireland and Sweden conceptualise technology education. While both nations share a commitment to technological scientific knowledge as a prerequisite foundation, their distribution of emphasis across the remaining legs, technical skills and socio-ethical understanding, reflects distinct national foci. The primary finding is a marked technical dominance within the Irish suite of technology subjects. The heavy focus on material manipulation and accuracy reinforces an identity centred on "knowing-how". In contrast, the Swedish curriculum, operating within the Didaktik tradition, treats technology as a social phenomenon. By dedicating nearly one-third of its core content to socio-ethical technical understanding.

This study identifies a significant "socio-ethical vacuum" in technically dense Irish subjects like Engineering and Graphics. The omission of social impact outcomes risks presenting technology as a value-neutral tool, potentially hindering the development of a holistic technological capability that accounts for human values. Furthermore, the structural delivery through a "suite model" in Ireland facilitates deep technical specialisation but may inadvertently create curricular silos, isolating technical competence from the critical frameworks required for ethical evaluation. These findings provide a critical insight that has the potential to inform the development of future technology subjects. To move beyond an insular reform trajectory, future specifications could integrate critical reviews utilising external frameworks and benchmarks.

Finally, it must be acknowledged that this analysis was confined to the intended curriculum. As the Didaktik tradition views teachers as autonomous "curriculum makers," much of the subject's depth may be realised in the enacted curriculum of the classroom rather than the written text. Future research should employ observational studies to determine if the theoretical breadth of the Swedish model and the technical density of the Irish model are fully mirrored in classroom practice.

Funding Open Access funding provided by the IReL Consortium

Declarations

Conflicts of Interest No perceived or actual conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material.

If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Banks, F., & Williams, P. J. (2022). International perspectives on technology education. In *Debates in design and technology education* (pp. 26–44). Routledge.

Barlex, D., & Steeg, T. (2017). *Re-Building Design & Technology In the secondary school curriculum* https://dandtfordandt.wordpress.com/wp-content/uploads/2017/05/re-building-dt-v21.pdf

Barry, C. A., Britten, N., Barber, N., Bradley, C., & Stevenson, F. (1999). Using reflexivity to optimize teamwork in qualitative research. *Qualitative Health Research, 9*(1), 26–44.

Beaumont, H., & Steeg, T. (2024). *Design and Technology in your school: Principles for curriculum, pedagogy and assessment*. Routledge.

Benson, C., & Lunt, J. (2011). *International handbook of primary technology education: Reviewing the past twenty years* (Vol. 7). Springer Science & Business Media.

Bladh, G. (2020). GeoCapabilities, Didaktical analysis and curriculum thinking – Furthering the dialogue between Didaktik and curriculum. *International Research in Geographical and Environmental Education, 29*(3), 206–220. https://doi.org/10.1080/10382046.2020.1749766

Cederqvist, A.-M. (2021). *Seeing the parts, understanding the whole-A technology education perspective on teaching and learning in processes of analysing and designing programmed technological solutions*. University of Gothenburg. https://gupea.ub.gu.se/server/api/core/bitstreams/f1d4cead-52cb-4814-a10e-a845ed3675c2/content

Clucas, P., & Sjöström, J. (2025). Traces of Bildung in upper secondary science education: A critical investigation of chemistry teachers' orientation towards promoting Bildung in chemistry education. *Science & Education, 34*(4), 2225–2250. https://doi.org/10.1007/s11191-024-00510-6

Dakers, J. R,. (Ed.). (2025). *A collection of dreams about the future of technology education: A Festschrift in honour of Marc J. de Vries (International Technology Education Studies,* Vol. 21). Brill. https://doi.org/10.1163/9789004710948

de Vries, M. J. (2005). The nature of technological knowledge: Philosophical reflections and educational consequences. *International Journal of Technology and Design Education, 15*(2), 149–154. https://doi.org/10.1007/s10798-005-8276-2

de Vries, M. J. (2016). *Teaching about technology: An introduction to the philosophy of technology for non-philosophers*. Springer.

de Vries, M. J. (Ed.). (2018). *Handbook of Technology Education* (1 ed.). Springer. https://doi.org/10.1007/978-3-319-44687-5.

Department of Education and Skills. (2018a). *Junior Cycle Applied Technology*. https://www.curriculumonline.ie/getmedia/2c3fc3c0-064c-4080-980e-a2738512b85b/Applied-Technology.pdf

Department of Education and Skills. (2018b). *Junior Cycle Engineering*. https://www.curriculumonline.ie/getmedia/80a6f0c8-e5a1-439e-a6af-cde49336f735/Engineering-Specification.pdf

Department of Education and Skills. (2018c). *Junior Cycle Graphics*. https://curriculumonline.ie/getmedia/ca152005-ae74-4360-914e-67da18f67474/Graphics.pd

Department of Education and Skills. (2018d). *Junior Cycle Wood Technology*. https://www.curriculumonline.ie/getmedia/805eea5b-22b8-47d3-9849-6530e500fe1d/Wood-Technology.pdf

DES. (2015). *Framework for Junior Cycle 2015*. https://ncca.ie/media/3249/framework-for-junior-cycle-2015-en.pdf

Doyle, A., Seery, N., & Gumaelius, L. (2019). Operationalising pedagogical content knowledge research in technology education: Considerations for methodological approaches to exploring enacted practice. *British Educational Research Journal, 45*(4), 755–769. https://doi.org/10.1002/berj.3524

Eurydice. (2022). *Informatics education at school in Europe*. Publications Office of the European Union. https://doi.org/10.2797/268406

Forman, J., & Damschroder, L. (2007). Qualitative content analysis. In L. Jacoby & L. A. Siminoff (Eds.), *Empirical methods for bioethics: A primer* (pp. 39–62). Emerald Group Publishing Limited. https://doi.org/10.1016/S1479-3709(07)11003-7

Fredriksson, U., Kusanagi, N. K., Gougoulakis, P., Matsuda, Y., & Kitamura, Y. (2020). A comparative study of curriculums for education for sustainable development (ESD) in Sweden and Japan. *Sustainability, 12*(3), Article 1123.

Gleeson, J., Doyle, A., & O'Neill, N. (2025). Student learning outcomes in the Irish context: mixing curriculum paradigms, cultures, and design models. *Irish Educational Studies*, 1–28. https://doi.org/10.1080/03323315.2025.2483241

Håkanson, L. (2010). The firm as an epistemic community: The knowledge-based view revisited. *Industrial and Corporate Change, 19*(6), 1801–1828. https://doi.org/10.1093/icc/dtq052

Hallström, J. (2025). Back to the Future: (Re) imagining a History-based Technology Curriculum. In J. R. Dakers (Ed.), *A Collection of Dreams about the Future of Technology Education.* Brill.

Hansson, S. O. (2013). What is technological knowledge? In *Technology teachers as researchers.* (pp. 17–31). Sense Publishers.

Hopmann, S. (2007). Restrained teaching: The Common Core of Didaktik. *European Educational Research Journal, 6*(2), 109–124. https://doi.org/10.2304/eerj.2007.6.2.109

Ihde, D. O. N. (2012). *Experimental phenomenology, second edition: Multistabilities*. State University of New York Press. https://doi.org/10.2307/jj.18253341

Jones, A., & de Vries, M. J. (2009). *International Handbook of Research and Development in Technology Education*. Brill. https://doi.org/10.1163/9789087908799

Jones, A., Buntting, C., & de Vries, M. J. (2013). The developing field of technology education: A review to look forward. *International Journal of Technology and Design Education, 23*, 191–212. https://doi.org/10.1007/s10798-011-9174-4

Kelly, P. (2014). Intercultural comparative research: Rethinking insider and outsider perspectives. *Oxford Review of Education, 40*(2), 246–265. https://doi.org/10.1080/03054985.2014.900484

Mathisen, Ø., & Johansen, G. (2025). Technological determinism? Exploring students' deliberations on renewable energy technology. *International Journal of Technology and Design Education, 35*(4), 1471–1491. https://doi.org/10.1007/s10798-025-10002-5

Mitcham, C. (1994). *Thinking through technology: The path between engineering and philosophy*. University of Chicago Press.

Moser, A., & Korstjens, I. (2018). Series: Practical guidance to qualitative research. Part 3: Sampling, data collection and analysis. *European Journal of General Practice, 24*(1), 9–18. https://doi.org/10.1080/13814788.2017.1375091

Mullis, I. V., Martin, M. O., Foy, P., Kelly, D. L., & Fishbein, B. (2020). TIMSS 2019 international results in mathematics and science. In: Boston College, TIMSS & PIRLS International Study Center.

NCCA. (2019). *Focus on Learning Outcomes: Workshop 5*. https://ncca.ie/media/4107/learning-outcomes-booklet_en.pdf

NCCA. (2024a). *Framework for Junior Cycle*. National Council for Curriculum Assessment. Retrieved 27/06/2024 from https://ncca.ie/en/junior-cycle/framework-for-junior-cycle/

NCCA. (2024b). Key competencies in senior cycle. https://ncca.ie/media/mfhhagys/key-competencies-in-senior-cycle_en.pdf

Nordlöf, C., Norström, P., Höst, G., & Hallström, J. (2022). Towards a three-part heuristic framework for technology education. *International Journal of Technology and Design Education, 32*(3), 1583–1604. https://doi.org/10.1007/s10798-021-09664-8

Norström, P. (2014). *Technological knowledge and technology education* KTH Royal Institute of Technology]. https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A715009&dswid=1585

Norström, P. (2022). Technological Systems in National Standards and Curricula. In J. Hallström & P. J. Williams (Eds.), *Teaching and Learning about Technological Systems: Philosophical, Curriculum and Classroom Perspectives* (pp. 35–51). Springer Nature Singapore. https://doi.org/10.1007/978-981-16-7719-9_3

Nye, D. E. (2006). *Technology Matters: Questions to Live with*. MIT Press. https://books.google.ie/books?id=dwbMsgEACAAJ

Phillips, D., & Schweisfurth, M. (2014). *Comparative and international education: An introduction to theory, method, and practice*. A&C Black.

Power, J. R., & Buckley, J. (2024). The purpose of second level engineering education: An examination of curricular consistency and industry alignment. *Australasian Journal of Engineering Education*. https://doi.org/10.1080/22054952.2024.2430155

Rasinen, A. (2003). An analysis of the technology education curriculum of six countries. *Journal of Technology Education, 15*(1), 31–47.

Ropohl, G. (1997). Knowledge types in technology. *International Journal of Technology and Design Education, 7*(1), 65–72. https://doi.org/10.1023/A:1008865104461

Schatzberg, E. (2018). *Technology: Critical History of a Concept*. University of Chicago Press. https://books.google.ie/books?id=DZ92DwAAQBAJ

Skolverket. (2022a). *Comments on the technology curriculum. Skolverket*. https://www.skolverket.se/getFile?file=9801

Skolverket. (2022b). *Curriculum for the Compulsory School, Preschool Class and School-Age Ecucare. Skolverket*. https://www.skolverket.se/getFile?file=13128

Sorsa, M. A., Kiikkala, I., & Åstedt-Kurki, P. (2015). Bracketing as a skill in conducting unstructured qualitative interviews. *Nurse Researcher*. https://doi.org/10.7748/nr.22.4.8.e1317

Spendlove, D. (2017). Design thinking: What is it and where might it reside? In E. Norman & K. Baynes (Eds.), *Design Epistemology and Curriculum Planning*. Loughborough Design Press Ltd. https://research.manchester.ac.uk/en/publications/design-thinking-what-is-it-and-where-might-it-reside

Stronati, C. (2023). *The design of upper secondary education across OECD countries: Managing choice, coherence and specialisation* (1993–9019). (OECD Education Working Papers, Issue. https://doi.org/10.1787/158101f0-en

Tahirsylaj, A., & Werler, T. C. (2021). In search of dimensions of bildung: A quantitative approach. *Nordic Studies in Education, 41*(1), 1–23. https://doi.org/10.23865/nse.v41.2130

Turja, L., Endepohls-Ulpe, M., & Chatoney, M. (2009). A conceptual framework for developing the curriculum and delivery of technology education in early childhood. *International Journal of Technology and Design Education, 19*(4), 353–365. https://doi.org/10.1007/s10798-009-9093-9

Vincenti, W. G. (1990). *What engineers know and how they know it*. Johns Hopkins University Press.

Wagner, J., & Hastedt, D. (2022). *Valuing Curriculum-Based International Large-Scale Assessments, Ensuring Alignment With National Curricula in IEA Studies.* (IEA Compass: Briefs in Education No. 16, Issue. IEA. https://www.iea.nl/sites/default/files/2022-03/IEA%20Compass%2016%20Valuing%20Curriculum-Based%20International%20Large-scale%20Assessments.pdf

Westerlund, S. (2023). A 'reverse'academic drift?: Changes in Swedish educational crafts. FORMakademisk,

Williams, P. J. (2015). Vocational and general technology education. In P. J. Williams, A. Jones, & C. Buntting (Eds.), *The future of technology education* (pp. 201–216). Springer. https://doi.org/10.1007/978-981-287-170-1_11

Winner, L. (2017). Do artifacts have politics? In J. Weckert (Ed.), *Computer ethics* (pp. 177–192). Routledge. https://doi.org/10.4324/9781315259697

Yip, S. Y. (2024). Positionality and reflexivity: Negotiating insider-outsider positions within and across cultures. *International Journal of Research & Method in Education, 47*(3), 222–232. https://doi.org/10.1080/1743727X.2023.2266375

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.